COURSE 2 · VIDEO B2 · EXERCISE

Total Target Cost measure

Video B2 · First steps: Power Query, model, DAX

Start: Engxtech_B2.pbix (your file from video B2); save a copy as Engxtech_B2_Exercise.pbix

Time: about 10 min

Data: the course data folder

DESIGN QUESTION

What does one unit of each product cost in parts, at target cost, and which part drives that cost?

Extra practice. This is the "try one thing on your own" task from the end of video B2 (Total Target Cost), written out with checks. Do it after the video, in a copy of your file.

In video B2 you wrote Total Target Mass with SUMX and RELATED. The same pattern works for cost: BOM holds the quantity per product, Parts holds the target cost of one part.

Your tasks

1. Open Engxtech_B2.pbix and save a copy as Engxtech_B2_Exercise.pbix (your B2 file stays the start of Section 2). In Power Query, check that Parts[TargetCost_USD] is a Decimal number.
2. Check the two relationships from the video: Parts[PartID] 1:* BOM[PartID] and Products[ProductID] 1:* BOM[ProductID].
3. On the BOM table, create the measure **Total Target Cost**: quantity per product times the target cost of the part, summed over BOM rows.
4. Format it as Currency (USD) with 2 decimals. Show it on a card next to Total Target Mass.
5. Add a table: Products[ProductName], Total Target Cost. Then a second table: Parts[PartName], Total Target Cost, sorted high to low.
6. Use the product slicer from B2 and read the cost of each product.

Check your answer

Check	Expected
Total Target Mass (control: should still read)	**14,865**
Total Target Cost, all products (USD)	**412.00**
Centrifugal pump (CP-100)	**129.50**
Booster pump set (BP-200)	**97.00**
Motorised control valve (MV-300)	**185.50**
Largest part line: Drive Motor (USD)	**170.00**

Hint. If every product shows the same number, a relationship is missing or its direction is wrong. If you get an error about a single value, you used a column without RELATED inside SUMX.

Stuck? The worked solution with the full DAX and M is in Solutions/B2_Solution.pdf.